Trust That Survives Deployment: Shift-Robust Review Guarantees for AI Medical Scribes

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Abstract

AI medical scribes write clinical notes from doctor-patient conversations, and their safety rests on a clinician reviewing each note. Conformal risk control (CRC) can tell that reviewer which sentences to check while bounding the expected share of errors left unflagged, but the bound assumes the notes at deployment look like the notes used to calibrate it. A scribe deployment breaks that assumption: vendors update models, microphones and accents change the transcript, and a hospital's cases differ from any benchmark.

We build **ScribeErr**, 11,996 corrupted notes from ACI-Bench and MTS-Dialog with 11 transcript-aware error types, including *grounded* errors in which every word was spoken during the visit but the meaning is wrong. We stress-test CRC review gates under eight deployment shifts, propose **type-conditional review guarantees** and a **label-free shift alarm** that keep the guarantee when the error mix changes, and audit 1,200 real notes from ten note writers.

Five findings:

1. **A valid guarantee can hide a blind spot.** In distribution, CRC meets its target (miss rate 0.098 at $\alpha = 0.10$) while missing 26.5% of grounded errors, mostly negation flips, and 0.4% of invented ones.
2. **A shift in error mix breaks the guarantee; transcript noise does not.** A vendor whose errors skew subtle pushes the miss rate to 0.155, violating the target in 97% of 300 resplits. ASR noise keeps it valid but raises reviewer workload from 28% to 41% of sentences.
3. **Type-conditional guarantees survive any error mix.** Bounding the miss rate inside each error family bounds it for every mixture of families. Across six conditions and 200 resplits it never exceeded 0.016, where marginal CRC failed in three conditions. Its price is set by the worst-detected family: 80% workload, almost entirely because of negation flips; without that family it costs the same as marginal CRC.

4. **A label-free alarm says when to pay that price.** Estimating the deployment error mix and error rate from unlabeled notes, the alarm fires in 94% to 100% of deployments with a harder error mix, transcript noise or a new corpus, in 1.5% when the shift makes errors easier, and in 4.5% when nothing has shifted. Switching to the type-conditional gate only when it fires keeps the miss rate at or below 0.025 under every shift while costing 30% workload when nothing has shifted.
5. **A verified report card on real outputs.** Transcript checks with 93% adjudicated precision find an invented drug, dose or side in 15% of GPT-4 notes against 1.7% of human reference notes, and a transcript-to-note certainty audit finds models turning the clinician's hedged diagnoses into certain ones where human notes did not (pilot, 23 diagnoses).

The result reframes the reviewer's question from "how often does the scribe err?" to "which errors am I guaranteed to see, and when has that stopped being true?" Code, labels, and a clinician labeling kit are released.

1. Introduction

Health systems are buying AI scribes faster than they can evaluate them. The assumed safety valve is the signing clinician, yet a recent census of three commercial scribes found a verified failure in about one note in three, including a failure mode no published taxonomy covered: a treatment the clinician retracted, recorded as delivered care (Fox et al., 2026). Researchers at Suki have argued that no existing tool combines sentence-level error detection, inter-rater reliability, and a statistical procedure for release decisions (Fierce Healthcare). Vendors meanwhile publish their own comparisons.

The statistical procedure now exists in one form. CARE (Bedi, Lin et al., 2026) applies conformal risk control (CRC; Angelopoulos et al., 2022) to medical summaries. It flags sentences so that the risk of an unflagged hallucination stays below a target, and it holds across 100 random resplits of five tasks. Like every conformal method, the guarantee assumes the calibration notes and the deployment notes are exchangeable. That assumption is exactly what a scribe deployment breaks.

A second line of work asks whether language models keep the clinician's uncertainty. "Possible or Definite?" (Du, Lu and Qu, MLHC 2026) finds that LLMs rewriting clinical documents preserve uncertainty cues less than half the time, almost always by collapsing hedges into definite statements. "From 'May' to 'Is'" finds the same bias toward inflated certainty in general rewriting. Neither study tests the scribe setting, where the source is a spoken conversation rather than a written note.

This paper connects the two. Our questions:

- **Q1.** Which kinds of scribe error can a sentence-level auditor see, and which are invisible to the grounding checks most auditors use?
- **Q2.** Does a CRC review guarantee calibrated on one distribution hold when the error mix, the transcript quality, or the corpus changes?
- **Q3.** If it breaks, how many labeled local notes restore it, and what does a shift-robust alternative cost in reviewer time?
- **Q4.** Can a guarantee survive error-mix shift with no deployment labels, and can a deployer tell from unlabeled notes when the calibrated guarantee has stopped holding?
- **Q5.** On real notes written by published systems, what do verified transcript checks and a certainty audit find?

Contributions. (i) ScribeErr, a transcript-aware error benchmark that separates *grounded* errors (all words spoken, meaning wrong) from *ungrounded* ones, with retraction and negation types. (ii) To our knowledge, the first stress test of CRC review guarantees under scribe-deployment shift, with a local recalibration curve. (iii) **Type-conditional review guarantees** that hold under any error mix without deployment labels, a trust dial between marginal and type-conditional guarantees, and the finding that their price is set by the worst-detected error family. (iv) A **label-free shift alarm** and an alarm-gated policy that pays for robustness only when the alarm fires, plus negation-aware evidence features that lower the price. (v) A reproducible report card on 1,200 real notes from ten note writers, with every flag on GPT-4 and human notes adjudicated, and a transcript-to-note certainty audit. (vi) A clinician labeling kit with an agreement statistic, so the report card can become a labeled corpus.

2. The ScribeErr benchmark

Sources. ACI-Bench (Yim et al., 2023) pairs 207 simulated primary-care and specialty visits with full clinical notes. Its virtscribe subset also provides a human transcription *and* a raw ASR transcript of the same audio, which gives us real transcript noise for 28 test encounters. MTS-Dialog (Ben Abacha et al., 2023) pairs shorter dialogues with single note sections and serves as an unseen corpus. Detectors are trained only on ACI-Bench train and validation encounters (87); all evaluation uses the 120 test encounters or MTS-Dialog.

Corruption. Each instance takes a reference note, leaves it clean with probability 0.2, and otherwise injects one to four errors. Every injection is checked against the transcript, so we know whether the corrupted sentence is still fully *grounded* (every content word

appears somewhere in the visit) or not.

Type	What changes	Grounded?
U_DOSE	dose changed to a value never said	no
U_NUM	vital, lab, duration or age changed to an unsaid value	no
U_MED	drug swapped for a same-class drug never mentioned	no
U_LAT	left/right flipped when the visit names one side only	no
U_TIME	time unit shifted (days to weeks)	no
U_FAB	fabricated symptom sentence inserted	no
NEG_FLIP	polarity reversed (denies to reports)	yes (words all said, polarity reversed)
G_NUM	number swapped for another number said in the visit	yes
G_MED	drug swapped for another drug discussed in the visit	yes
G_LAT	side flipped when the visit mentions both sides	yes
RETRACT	transcript edited so the clinician retracts a plan the note still records	yes

Retractions use two disjoint phrasing sets: set A ("let's hold off on the {drug}") for training and set B ("I'm going to take back the {drug}") for every test condition, so a detector cannot pass by memorizing phrases. Severity is assigned by rule: drug, dose and retraction errors are major; number and side errors are major in plan, medication, results and exam sections and moderate elsewhere. In the test set, 57% of injected errors are major.

Splits and shift conditions.

Split	Notes	Sentences	Errors
Train (87 encounters x 20 seeds)	1,740	60,980	2,873
Test (120 encounters x 5 seeds x 3 mixes x 5 ASR rates)	9,000	322,530	15,345
Paired real ASR (28 encounters x 10 seeds x 2 transcripts)	560	20,678	970
MTS-Dialog (102 dialogues)	696	4,672	853

The test set crosses three **error mixes** with five **ASR rates**. The *uniform* mix weights types equally (42% of errors grounded). A *subtle vendor* mix models a mature scribe that rarely invents facts but mis-binds numbers and drugs and misses retractions (82% grounded). A *crude vendor* mix models an early scribe that mostly invents things (12% grounded). Simulated ASR noise applies word substitutions, deletions and insertions at 5% to 30%, over-weighting sound-alike drugs, teen/ty number confusions and dropped laterality. The paired virtscribe set replaces simulation with real ASR output.

3. The auditor and the review gate

Evidence features. For every note sentence the auditor computes about 40 features against the transcript, in six groups: *retrieval* (word and character TF-IDF similarity to the best three-turn window), *support* (share of content words, numbers, drugs and symptoms absent from the visit or from the matched window), *binding* (is a dose written next to a drug in the note also near that drug in the visit?), *laterality* (is the stated side in the matched window, or only the opposite side?), *polarity* (note negation against the window's polarity, including doctor-question and patient-answer pairs), and *reversal* (was the drug's last mention in the visit a stop, hold or cancel turn?). Section indicators are included.

Detectors. Two baselines need no training: *lexical* (share of content words never said, the grounding heuristic most auditors use) and *similarity* (one minus the best window similarity). Three trained detectors use the features: logistic regression (SC-LR), gradient-boosted trees (SC), and SC with the reversal group (SC+R). Out-of-fold isotonic regression on the training set provides calibrated probabilities. Nothing is tuned on test data.

Loss and guarantee. A sentence goes to the reviewer when its score is at least λ. For a note with erroneous sentences, the loss is the fraction of them left unflagged; a note with no errors has loss 0. CRC picks the largest λ (least review) with $(n \cdot \hat{R}(\lambda) + 1) / (n + 1) \leqslant \alpha$ on n calibration notes, which guarantees expected loss at most α on an exchangeable new note. We compare four gates:

- **CRC**: the standard marginal guarantee above.
- **CRC-sev**: the same, with each error weighted by severity (major 1, moderate 0.5, minor 0.25).
- **CRC-HP**: a high-probability version using a Hoeffding upper bound, $\hat{R}(\lambda) + \sqrt{\log(1/\delta) / 2n} \leqslant \alpha$ with $\delta = 0.1$, so the guarantee holds with 90% confidence over the calibration draw.
- **Strat-CRC**: separate thresholds inside a high-risk stratum (plan, medication, allergy and results sections, or any sentence naming a drug or dose) and outside it, giving a per-stratum guarantee.

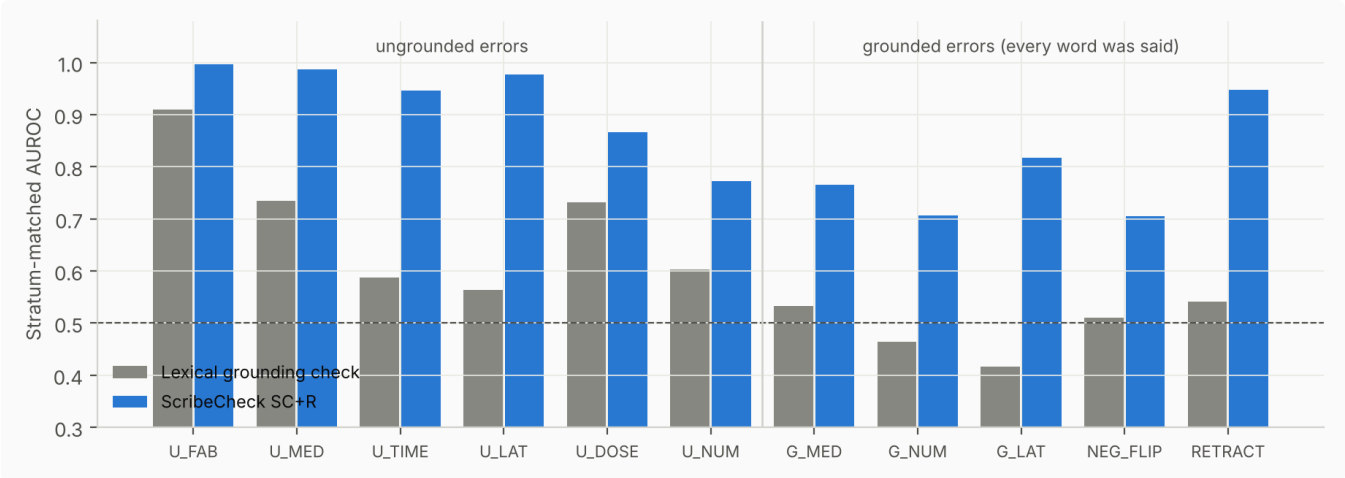
Protocol. In each of 300 resplits, half of the 120 test encounters are drawn for calibration on the in-distribution condition (uniform mix, clean transcript). The gates are then applied to the held-out encounters under each shift; MTS-Dialog uses all its dialogues because it shares no encounters. The paired real-ASR experiment calibrates on human transcripts and deploys on raw ASR of held-out encounters. We report the mean realized loss, reviewer workload (share of sentences flagged), the share of resplits whose realized loss exceeds α, and miss rates by error type. We use $\alpha = 0.10$ in the text; results at 0.05 and 0.20 are in the released tables.

4. Results on ScribeErr

4.1 What the auditor can and cannot see (Q1)

Detector	AUROC	AUPRC	ECE (calibrated)
Lexical grounding	0.587	0.062	–
Similarity	0.554	0.060	–
SC-LR	0.852	0.396	0.004
SC	0.905	0.500	0.002
SC+R	0.913	0.533	0.002

The lexical check, the standard grounding heuristic, is near chance overall (0.59). On grounded error types it is *below* chance (stratum-matched AUROC 0.42 to 0.53), because a cross-bound dose or a flipped side shares more words with the visit than an ordinary sentence does. To stop section and drug base rates from doing the work, we compare each error only against clean sentences from the same stratum (section x mentions a drug x has a number). On that harder test, SC+R reaches 0.87 to 1.00 on ungrounded types, but only 0.71 on grounded numbers, 0.71 on negation flips, and 0.77 on grounded drugs. The reversal features lift retraction detection from 0.60 (SC) to 0.95 (SC+R), even though training and test use disjoint retraction phrasings.



4.2 Does the guarantee survive deployment? (Q2)

Realized miss rate and workload at α = 0.10, SC detector, mean over 300 resplits:

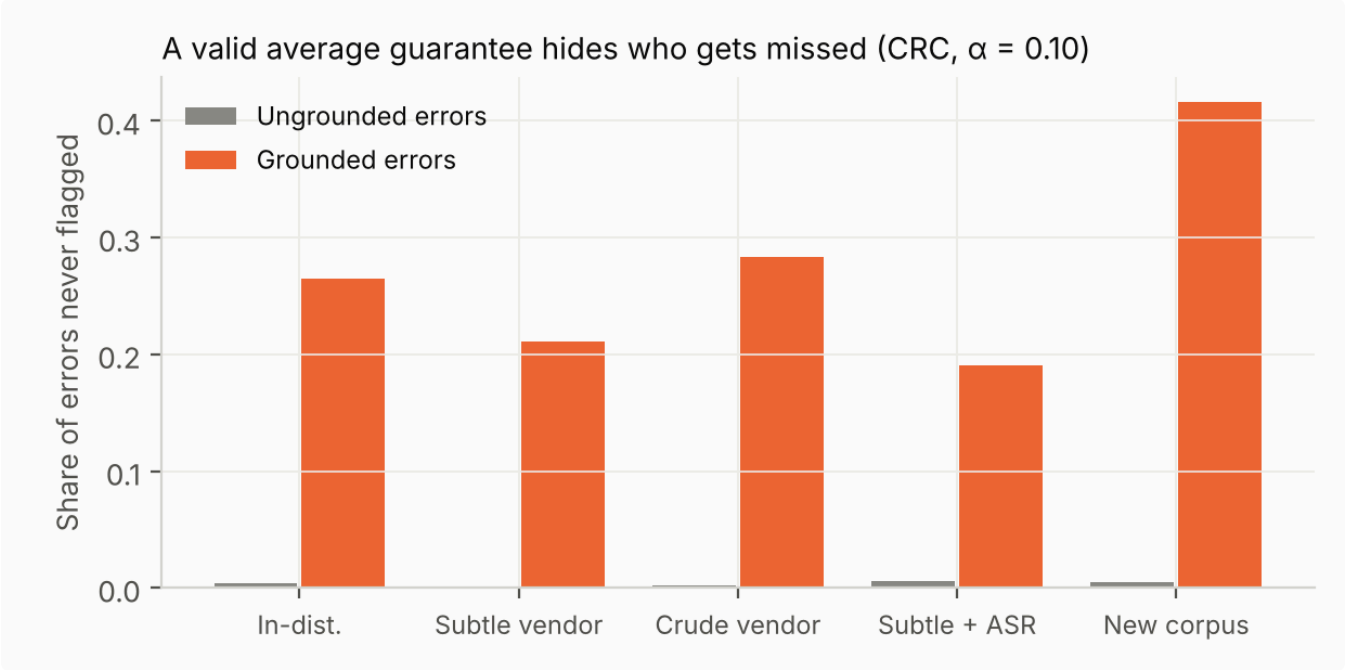
Deployment condition	CRC miss rate	P(miss > α)	CRC workload	CRC-HP miss rate	CRC-HP workload
In-distribution	0.098	0.44	28%	0.040	52%
Simulated ASR 10%	0.094	0.34	30%	0.040	54%
Simulated ASR 30%	0.085	0.22	35%	0.041	57%
Real ASR (paired)	0.078	0.24	41%	0.005	91%
New corpus (MTS-Dialog)	**0.107**	**0.79**	36%	0.029	62%
Crude vendor mix	0.023	0.00	28%	0.008	52%
Subtle vendor mix	**0.155**	**0.97**	28%	0.055	52%
Subtle vendor + ASR 20%	**0.136**	**0.91**	31%	0.046	55%

Two kinds of shift behave in opposite ways. Transcript noise makes clean sentences look less supported, so more sentences cross the threshold. The miss rate *falls* and the cost appears as reviewer workload, which rises from 28% to 41% of sentences on real ASR. Error-mix shift does the reverse. It leaves the score distribution of clean sentences unchanged, so the gate flags the same share of sentences, but the errors it now faces are the kind it scores lowest. A subtle vendor pushes the miss rate 55% over target, and the guarantee fails in 97% of resplits. Note that P(miss > α) near 0.4 in distribution is expected: CRC bounds the *mean* over calibration draws, not every draw.



4.3 A valid average can hide who is missed

In distribution, where CRC holds, it misses **26.5%** of grounded errors and **0.4%** of ungrounded ones, a 66-fold gap. By type: negation flips 61%, grounded side flips 28%, grounded numbers 16%, retractions 0.1% (retracted plans sit in drug-bearing plan sentences, which score high regardless), invented doses and fabricated symptoms 0%. Negation flips make up 32% of grounded errors but about three-quarters of the grounded misses, so the blind spot is mostly a negation blind spot. The marginal guarantee is met by catching nearly every easy error. Stratified CRC does what it promises inside the high-risk stratum in distribution (0.099), but under the subtle-vendor shift that stratum's miss rate reaches 0.187. Severity weighting barely moves the threshold, because hard errors are not concentrated in low-severity classes.



This matters because grounded errors are what a mature scribe makes. The census of commercial scribes cited above independently reports the retraction pattern, and grounded errors are also the hardest for a human reviewer: the wrong sentence reads as perfectly ordinary.

4.4 Repair: local recalibration and its price (Q3)

Recalibrating CRC on m labeled notes from the deployment site restores the target under all three breaking shifts. At m = 80 the mean miss rate is 0.091 to 0.094 with 40% workload; at m = 10 the finite-sample correction makes the gate conservative (miss rate about 0.05) at a cost of 61% to 64% workload. CRC-HP needs no local labels and stays

under target in every condition, at roughly twice the in-distribution workload. The practical rule: ship CRC-HP until about 80 locally labeled notes exist, then switch to a locally calibrated CRC.



5. Making the guarantee survive shift without local labels (Q4)

Section 4 leaves a deployer with two bad options: label about 80 local notes first, or accept CRC-HP's doubled workload forever. Neither can say *when* the calibrated guarantee stops being trustworthy. We propose three pieces that together make the guarantee robust to error-mix shift and tell the deployer when to pay for that robustness.

5.1 Type-conditional review guarantees

Group errors into three families: *invented* (all U_ types), *negation* (NEG_FLIP) and *grounded binding* (G_NUM, G_MED, G_LAT, RETRACT). Error-mix shift changes how often each family occurs, but within a family an error looks the same to the auditor; this is the label-shift assumption. **Type-conditional CRC (TC-CRC)** runs the conformal correction separately inside each family and uses the strictest resulting threshold. Every family's miss rate is then at most α, so the miss rate of *any* mixture of families is at most α. The guarantee needs no deployment labels and no estimate of the deployment mix, and by construction it closes the blind spot of Section 4.3.

The trust dial. Between marginal CRC and TC-CRC we define MR-CRC(h): the miss rate is guaranteed for every error mix whose hard share (negation plus grounded) is at most h, by requiring $(1 - h) \cdot \text{miss_invented} + h \cdot \max(\text{miss_negation}, \text{miss_grounded}) \leqslant \alpha$ with finite-sample corrections. A buyer sets h from what they are willing to assume about a vendor.

5.2 A label-free shift alarm

A four-way family classifier (no error and the three families), trained on ScribeErr-train, scores every sentence of the deployment notes. Its probabilities are bias-corrected on the labeled calibration notes, and the deployment family proportions are re-estimated from unlabeled notes by expectation maximization (Saerens et al., 2002; Alexandari et al., 2020). On clean transcripts the estimate is accurate: hard share 0.44 estimated against 0.42 true with no shift, and 0.83 against 0.82 for the subtle vendor. Transcript noise biases it, but noise also roughly doubles the estimated error rate, so the alarm uses two signals: it fires when either the estimated hard share or the estimated error rate leaves its in-distribution range (each threshold at the 97.5th percentile of in-distribution draws). The **alarm-gated policy** runs cheap marginal CRC and switches to TC-CRC only when the alarm fires.

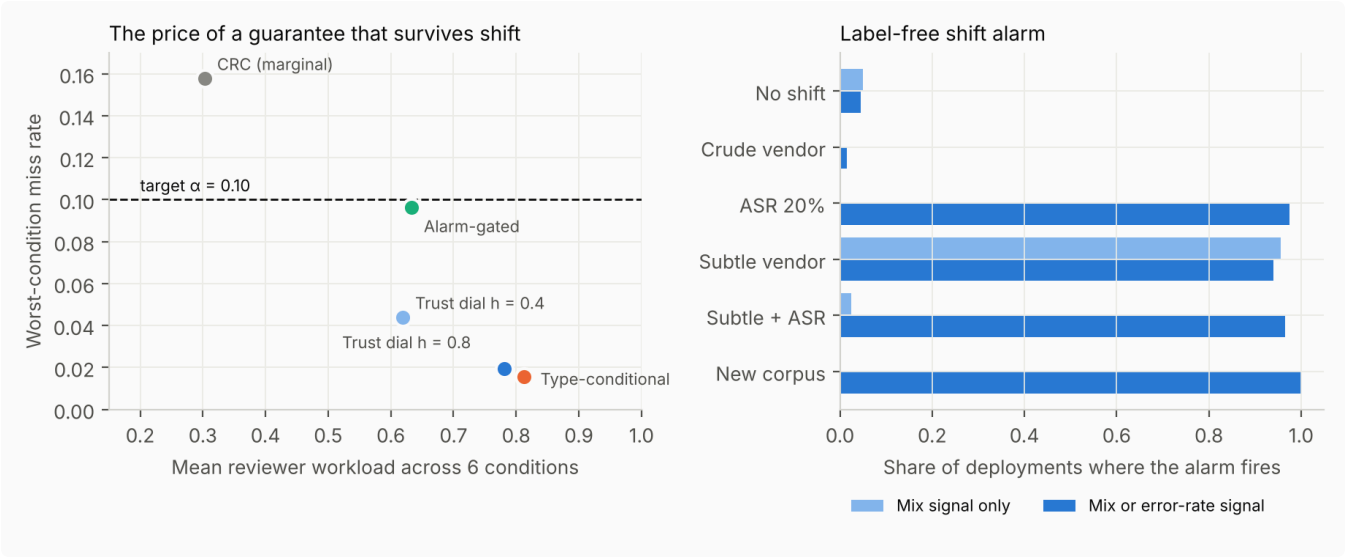
5.3 Results

SC auditor, $\alpha = 0.10$, 200 resplits. Each cell shows mean miss rate / reviewer workload.

Gate	No shift	Crude vendor	ASR 20%	Subtle vendor	Subtle + ASR	New corpus
CRC (marginal)	0.100 / 28%	0.023 / 28%	0.091 / 32%	**0.158** / 27%	**0.138** / 31%	**0.109** / 36%
Trust dial, h = 0.4	0.030 / 60%	0.005 / 60%	0.031 / 62%	0.044 / 59%	0.035 / 62%	0.020 / 69%
TC-CRC	0.015 / 80%	0.000 / 80%	0.015 / 81%	0.016 / 80%	0.015 / 81%	0.002 / 84%
Alarm-gated	0.096 / 30%	0.023 / 29%	0.017 / 80%	0.025 / 78%	0.019 / 80%	0.002 / 84%
Alarm fire rate	4.5%	1.5%	97.5%	94%	96.5%	100%

TC-CRC and every setting of the trust dial hold the target in all six conditions and all 200 resplits; marginal CRC fails in three conditions, in 82% to 98% of resplits. The alarm-gated policy keeps marginal CRC's cost when nothing has shifted (30% workload, within 2 points of CRC) and the type-conditional guarantee when something has. Its residual risk is the 6% of subtle-vendor deployments where the alarm stays silent.

The price of trust is set by the worst-detected error family. TC-CRC costs 80% workload because negation flips are nearly invisible to the auditor. Drop the negation family and the type-conditional gate costs the same as marginal CRC (29% workload). So guaranteeing that grounded binding errors are caught is almost free; guaranteeing negation is not.



5.4 Lowering the price: negation-aware evidence

The auditor's polarity features look at the best-matching window, which "denies chest pain" and "endorses chest pain" retrieve equally. We add *term-anchored polarity*: every clinical term in a note sentence is traced to every visit turn that mentions it, and each mention is scored as affirmed, denied (a negation cue before it, or a "no" answer to the doctor's question about it), or neutral. The features ignore grammar on purpose, so the auditor cannot learn the wording artifacts of our injected negations. Negation-flip detection rises from 0.69 to 0.74 stratum-matched AUROC (overall AUROC 0.905 to 0.912). With retraction and polarity features combined (SC+RN), marginal CRC's workload falls from 28% to 22% and TC-CRC's from 80% to 75%, with all guarantees intact. Negation remains the open problem, and any better negation detector lowers the cost of a type-conditional guarantee directly.

6. Report card on real note writers (Q5)

ACI-Bench released the notes that nine systems produced for its 120 test encounters: four OpenAI models prompted zero-shot (GPT-4, ChatGPT, Text-Davinci-003 and -002) and five fine-tuned summarizers (BART-large, BART-xsum-samsum, BioBART, LED, LED-pubmed). With the human reference notes that makes ten note writers on the same visits, 1,200 notes in all. These are 2023-era systems, not today's commercial scribes; the point is a method that grades any writer the same way.

6.1 Verified hard checks

We use four deterministic checks whose flags anyone can verify in seconds: a drug the visit never mentions (brand and generic names merged), a dose number never said, a dose said but never near that drug, and a side stated when the visit only names the opposite side for a body part. Transcript normalization handles doses spoken aloud ("two point five", "seven and a half", "five hundred") and dose-before-drug order.

We adjudicated every flag on GPT-4 and human notes, plus a sample of 41 flags from the other systems, against the transcript (82 flags). Each flag was labeled *true error* (the visit does not support the claim), *transcript error* (the note is probably right but the ASR transcript mis-heard it, as in "57 twice a day" for 750 mg), or *false alarm*.

Source of flags	Flags	True error	Transcript error	False alarm
GPT-4 (all)	29	26	2	1
Human reference (all)	12	2	5	5
Other 8 systems (sample)	41	40	1	0
Total	**82**	**68 (83%)**	**8 (10%)**	**6 (7%)**

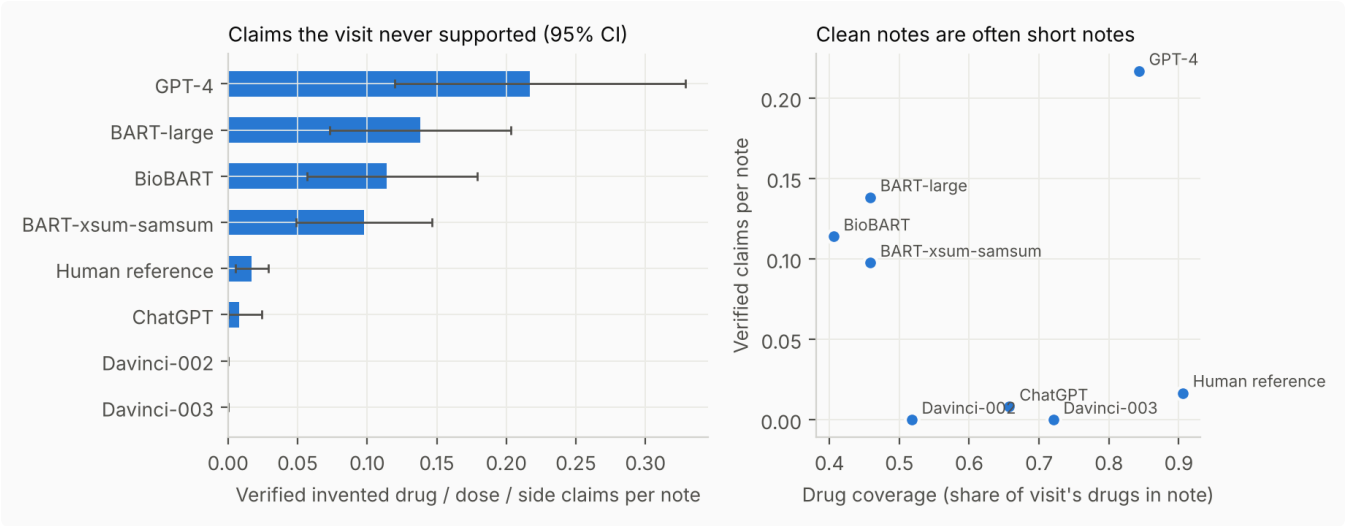
The checks are precise: 93% of flags mark content the transcript does not support, and 83% are genuine note errors. The contrast between sources is the finding. **18 of 120 GPT-4 notes (15%) contain at least one verified invented drug, dose or side, compared with 2 of 120 human reference notes (1.7%).** Most flags on human notes trace to garbled transcripts, not to the note.

The GPT-4 errors are fluent and clinically plausible, which is what makes them dangerous:

- A patient says she manages depression with "the medication you prescribed me"; the note names escitalopram 10 mg.
- A visit with no mention of antiplatelets gets "Continue aspirin 81 mg daily and clopidogrel 75 mg daily".
- An in-office chest-pain episode is recorded as "given aspirin, nitroglycerin, morphine, and heparin"; only the first two were mentioned.
- "You're on the high dose of atorvastatin" becomes "atorvastatin 80 mg"; "a very tiny dose of hydrochlorothiazide" becomes "12.5 mg".
- A right-knee visit gets the exam line "Left knee is normal."

The fine-tuned models fail more crudely ("metformin 5,000 mg", "Tylenol 2 mg every 6 to 8 hours"), and BioBART shows a textbook grounded error: the 15 mg in "Miralax 15 mg" was the dose the doctor gave for meloxicam.

A low flag count is not the same as a safe note. The Davinci models produce no flags, but they also leave out many of the visit's drugs: they record 52% to 72% of the drugs discussed, against 84% for GPT-4 and 91% for the human notes. The BART-family models score worst on both axes. A report card that ranks only on unsupported claims rewards a scribe that says less, so ScribeCheck always reports coverage beside it.

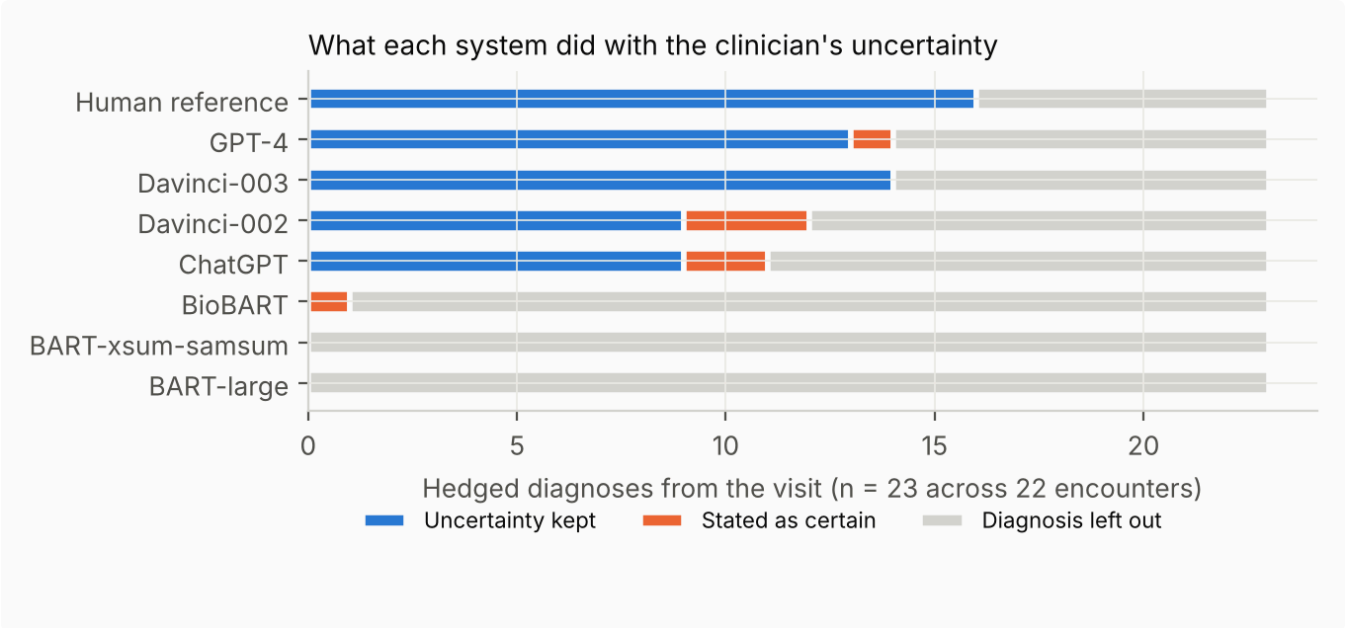


6.2 Certainty audit

For each visit we find diagnoses the clinician hedged with a strong uncertainty cue ("probably", "most likely", "rule out", "I suspect", "could be"), using a fixed lexicon of common diagnoses. Weak markers such as "I think you have" and "sounds like" are excluded, because clinicians routinely use them to state a working diagnosis. We then check how each note records the diagnosis where it states a diagnosis (assessment, plan or chief complaint, or any sentence with a diagnostic verb): with an uncertainty cue (*kept*), without one (*stated as certain*), or not at all (*left out*). Three rounds of manual validation shaped these rules. In round two, 34 of 37 status assignments on usable notes were correct; the errors (a nodule, a murmur, reflux as a symptom) led us to remove observed findings from the lexicon. Round three removed recurrent conditions ("recurrent strep") and added "appears" as a note-side cue. After that, every remaining *stated as certain* case on the eight usable writers (7 of 7) was correct on reading. The LED models' notes are largely incoherent and are excluded from validation.

Writer	Kept	Stated as certain	Left out
Human reference	16	0	7
Davinci-003	14	0	9
GPT-4	13	1	9
ChatGPT	9	2	12
Davinci-002	9	3	11
BioBART	0	1	22
BART-large	0	0	23
BART-xsum-samsum	0	0	23

ACI-Bench yields only 23 strongly hedged diagnoses across 22 visits, so these counts are a pilot and the intervals are wide. The pattern is still informative. Human scribes kept every hedge they recorded (16 of 16). Davinci-002 stated 3 of its 12 recorded hedged diagnoses as certain and ChatGPT 2 of 11: "your symptoms are most likely due to rotator cuff tendinopathy" became "The doctor diagnosed Betty with rotator cuff tendinopathy", and "I suspect that she likely has GERD" became "King has been diagnosed with gastroesophageal reflux disease". GPT-4 did this once. The fine-tuned summarizers rarely record the hedged diagnosis at all, an omission that no check for unsupported claims can see.



7. Limitations and next steps

- **Injected errors are not natural errors.** ScribeErr controls the error mix so that shift can be studied, but a real scribe's mix is unknown. The subtle and crude mixes bracket plausible vendors; they do not estimate any vendor. The real-output report card in Section 6 is the check on this, and the census of commercial scribes cited in Section 1 independently found the retraction and medication patterns we inject.
- **No clinician labels yet.** Every adjudication in Section 6 was done against the transcript by the authors with AI assistance, not by clinicians, and severity was assigned by rule. Agreement statistics (Krippendorff's alpha) can only come from independent raters. The labeling kit (Section 8) exists for exactly this step.
- **Old systems, simulated visits.** ACI-Bench encounters are role-played, and the generated notes come from 2023-era models used without scribe-specific prompting. The method transfers to current products; the numbers do not.
- **Narrow hard checks.** Drug, dose and side checks are precise because they are narrow. They cannot see a wrong diagnosis, an invented exam finding without a side, or an omission. The trained auditor covers more, but its calibration was learned on injected errors.
- **Small certainty sample.** 23 hedged diagnoses support a protocol and a pilot comparison, not a ranking.
- **CRC averages over notes.** All guarantees here are marginal (in expectation over notes and calibration draws) or, for CRC-HP, high-probability over calibration draws. None bound the miss rate on a single note or a single patient.
  Label-shift assumption. Type-conditional guarantees assume an error family looks the same to the auditor at every site. Transcript noise and a new corpus change that too; TC-CRC still held under both here, but nothing in the method guarantees it. Within-family guarantees are sentence-level and treat errors in the same note as exchangeable.
  Alarm blind spots. The alarm is tuned to a 5% false-alarm rate on in-distribution draws and missed 6% of subtle-vendor deployments, where the policy fell back to a gate that does not hold. Its mix estimate is biased under transcript noise; only the error-rate signal catches those cases.
  Injection wording. Some injected negation flips read unnaturally ("endorses any chest pain"). The negation-aware features ignore grammar so that they cannot exploit this, but detection rates on natural negation errors may differ.

Ethics. All data are public, de-identified or simulated. No real patient notes were used. A public report card on named commercial products would need the products' outputs under a documented protocol, and vendors should be able to reproduce every number; that is why every flag is released with its transcript evidence.

Next steps. (1) Run the labeling kit with three or more clinicians or senior medical students to obtain severity labels and an agreement estimate. (2) Generate notes from current models and, where accessible, commercial scribes on the same public visits. (3) Replace the deterministic certainty audit with a model-based one validated against the clinician labels, and scale it to longer visits where clinicians hedge more often.

8. Reproducibility

`./run_all.sh` rebuilds every number and figure from the public datasets in about 35 minutes on two CPU cores (Python 3, scikit-learn, pandas, matplotlib; no GPU and no API keys). The pipeline is `01_build_benchmark` (ScribeErr), `02_featurize`, `03_train_detector`, `04_experiments` (300 resplits), `05_real_scribes` (report card), and `07_figures`. Manual adjudications are released as `results/hardcheck_labels.csv`, with a one-line rationale per flag. `08_labeling_kit.py build` produces blinded, shuffled sheets for three raters (352 items: every hard-check flag, every hedged-diagnosis claim, and a score-weighted sample of unflagged sentences, so the corpus also measures misses). `08_labeling_kit.py score` reports Krippendorff's alpha per field. The rubric is in `labeling/RUBRIC.md`.

Section 5 runs from `10_polarity_features` (term-anchored polarity features and the SC+N and SC+RN auditors) and `09_shift_aware 200 SC` (type-conditional gates, the trust dial, the alarm and the alarm-gated policy; about 10 minutes). Results are in `results/mixrobust_summary_SC.csv` and `results/shift_alarm_SC.csv`, with `_SC+RN` versions for the negation-aware auditor.

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